Sahaj Kishorkumar Mistry

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Education

Indian Institute of Technology Delhi 2026 – 2028
Master of Technology in Machine Intelligence and Data Science

Indian Institute of Technology Jammu 2020 – 2024
Bachelor of Technology in Computer Science and Engineering; CGPA: 8.91/10

Experience

Samsung R&D Institute, Noida Sep 2025 – Jul 2026
R&D Engineer

- Built an on-device image captioning system (ViT + SLM) for Snapdragon 8 Gen 3 as an accessibility solution for visually impaired users.
- Applied INT8 quantization to reduce model memory from 1 GB to 250 MB and latency from ∼500 ms to ∼150 ms (4× memory reduction, 3.3× speedup).
- Developed camera primitive skills using LangChain and LangGraph for an AI agent to autonomously control native Samsung Android camera functionalities.

Miko, Mumbai Jun 2024 – Sep 2025
Applied Scientist (Junior AI Image Processing Engineer)

- Engineered a Generative Story Pipeline using LangChain with a Content Safety Module for robust guardrails; deployed on Cloud Run to support 500+ rpm.
- Developed a face-swapping model using controlled image-guided inpainting for generative content.
- Optimized model serving by containerizing Triton Server on GKE, reducing inference latency and cost by 30%.

Research Experience

University of Pennsylvania (Remote) May 2023 – Aug 2023
Researcher – Brain Tumor Segmentation

- Proposed a novel dual-encoder nnU-Net with cross-attention and ODConv3D layers for brain tumor segmentation.
- Achieved **1st rank** in BraTS Africa Challenge (Validation Phase); Dice scores of 0.835/0.848/0.756 (ET/TC/WT). Published at **MICCAI 2023**.

KAUST (Remote) May 2023 – Jul 2023
Research Intern – Panoramic Depth Estimation

- Adapted ZoeDepth for 360° panorama depth estimation; replaced convolutions with Omni-Dimensional Dynamic Convolution and trained LoRA adapters. Achieved 0.250 RMSE, surpassing SOTA.

IIT Jammu (under Prof. Vinit Jakhetiya) May 2022 – Apr 2024
Undergraduate Researcher

- **HR Depth Estimation (CVPR 2024):** Fine-tuned DepthAnything with multi-loss strategy for transparent/mirror surfaces; placed 7th.
- **Stereo Image Super-Resolution (CVPR 2023 & 2024):** Retrained NAFSSR with perceptual and texture losses; placed 7th and 13th.
- **Drone Detection (ICASSP 2023):** YOLOv7 + CSRT tracker pipeline; **2nd rank** with 0.89 validation mAP. Published at **IEEE ICASSP 2023**.

Selected Publications

MICCAI 2023 – Multiscale Encoder and ODConv Enrichment in nnU-Net for Brain Tumor Segmentation
IEEE ICASSP 2023 – Drone-vs-Bird: Drone Detection Using YOLOv7 with CSRT Tracker
CVPR 2024 – NTIRE Challenge on HR Depth from Images of Specular and Transparent Surfaces
CVPR 2023 – NTIRE Challenge on Stereo Image Super-Resolution: Methods and Results

Technical Skills

Domains: Multimodal LLMs, Agentic AI (Tool Use, Planning, Reflection), Diffusion Models, GNNs, CNNs
Frameworks: LangChain, LangGraph, CrewAI, AutoGen, LlamaIndex, Hugging Face Transformers, PyTorch
ML Engineering: Model Quantization (INT8), ONNX, TFLite, TensorRT, Triton Inference Server, FAISS
Languages & Tools: Python, C/C++, JavaScript, Docker, GKE, Cloud Run, FastAPI